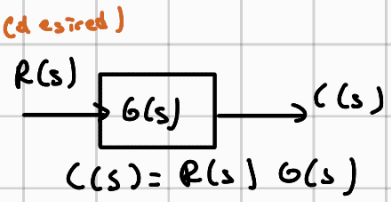
Representation methods in the Frey domain

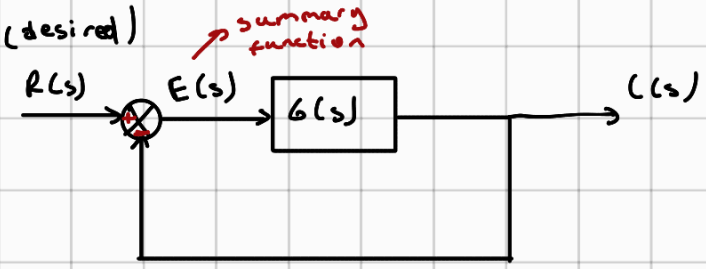
1- Block diagram ✓

2- Signal -flow plot

Transfer function (G(s)) = $\frac{\text{Output}}{\text{Input}} = \frac{C(s)}{R(s)}$



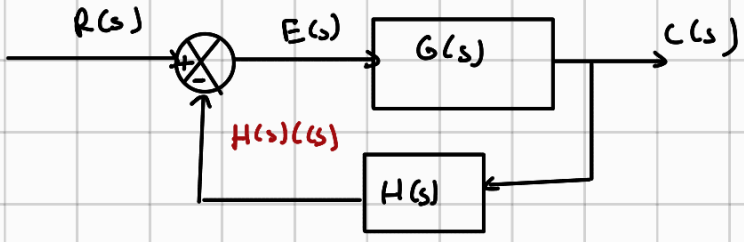
$C(s) = R(s)\,G(s)$



closed loop systems

$E(s) = R(s) - C(s)$

Transducers are required



Feed forward T.F.

F.F. = G(s)

Closed loop T.F

$\frac{C(s)}{R(s)}$

$E(s) = R(s) - H(s)\,C(s)$

$C(s) = E(s)\,G(s)$

$C(s) = [R(s) - H(s)\,C(s)]\,G(s)$

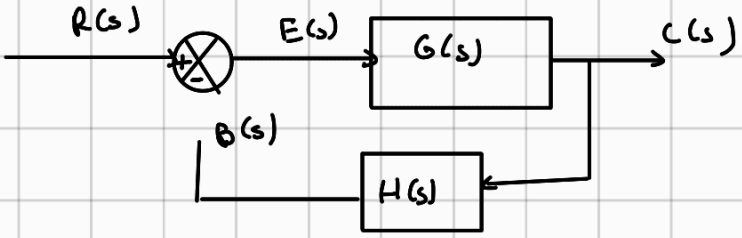
$C = RG - HCG$

$C[1 + HG] = RG$

$\frac{C}{R} = \frac{G}{1+HG} = \frac{\text{feed forward T.F}}{1 \mp \text{open loop T.F}}$

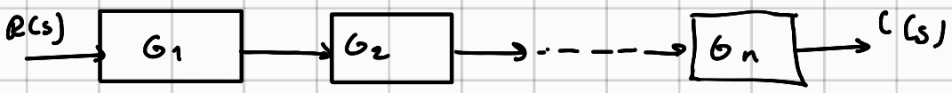
$\mp$ → for negative feedback

open Loop T.F.



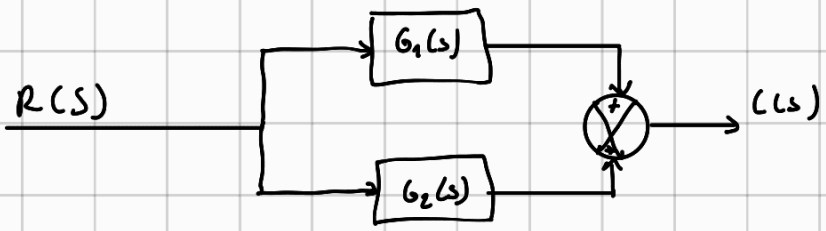
$$\frac{B(s)}{R(s)} = G(s)\,H(s)$$

Cascade configuration



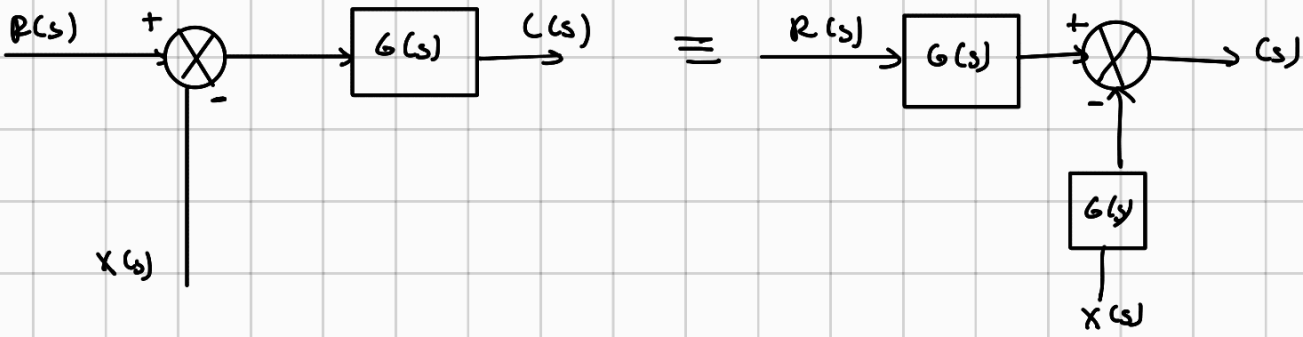
Cascade config

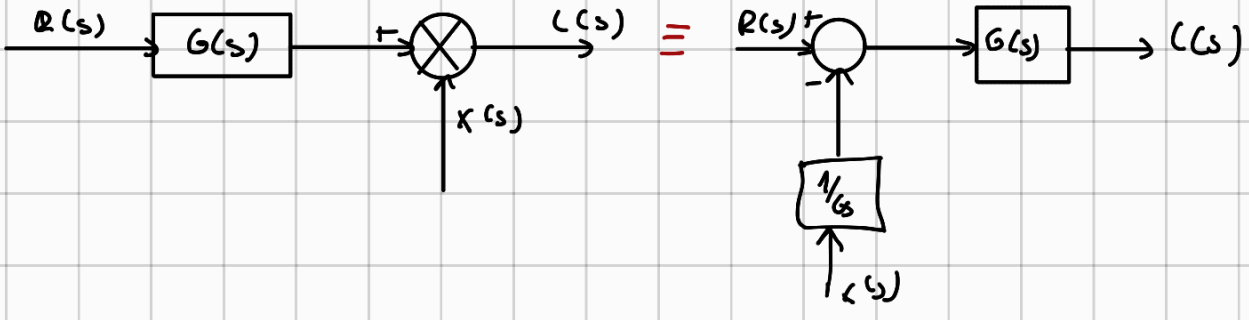
$$\frac{C(s)}{R(s)} = G_1\,G_2 \cdots G_n$$



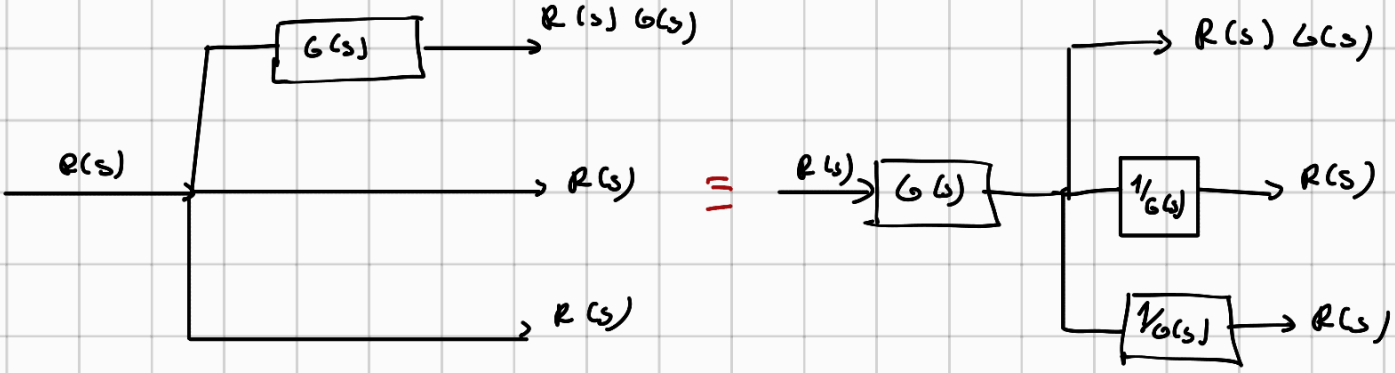
$$C(s) = R(s)\,G_1(s) + R(s)\,G_2(s)$$

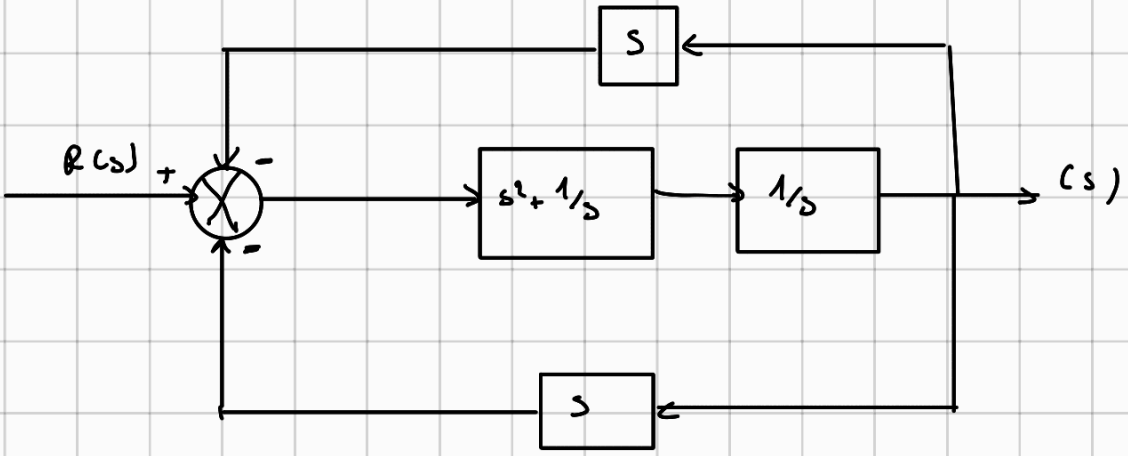
$$\frac{C(s)}{R(s)} = G_1(s) + G_2(s)$$



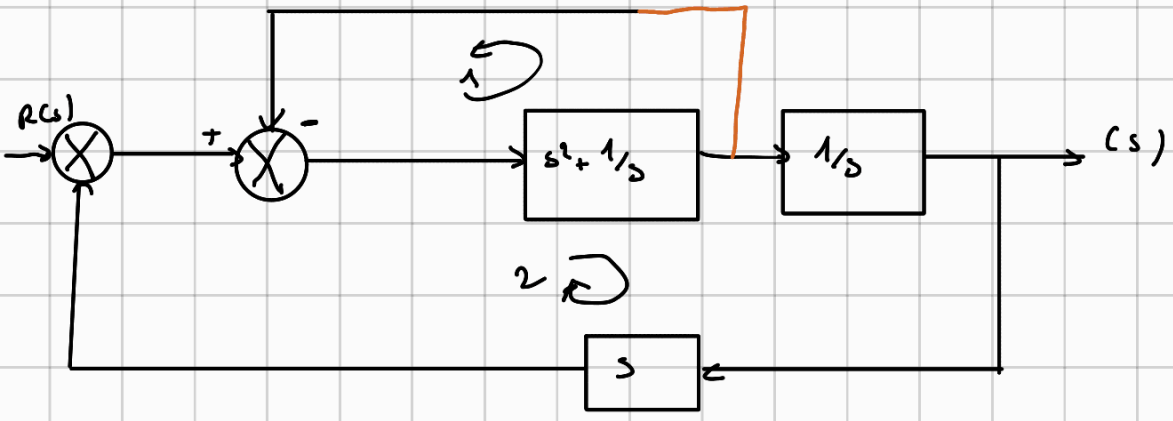


Pickoff points

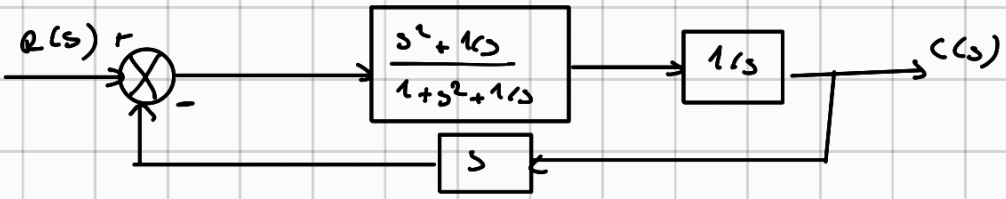




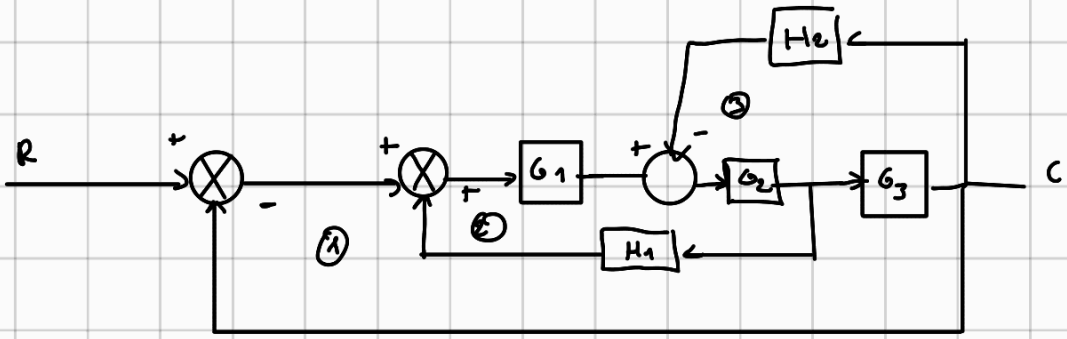
≡

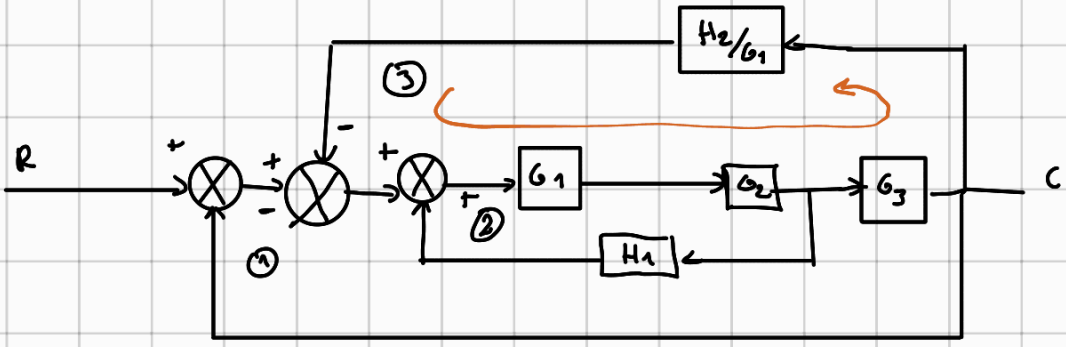


$$TF_1 = \frac{s^2 + 1/5}{1 + s^2 + 1/5}$$



$$\frac{C(s)}{R(s)} = \frac{(1/5)\left(\frac{s^2+1/5}{1+s^2+1/5}\right)}{1 + (1/5)\left(\frac{s^2+1/5}{1+s^2+1/5}\right) 5}$$





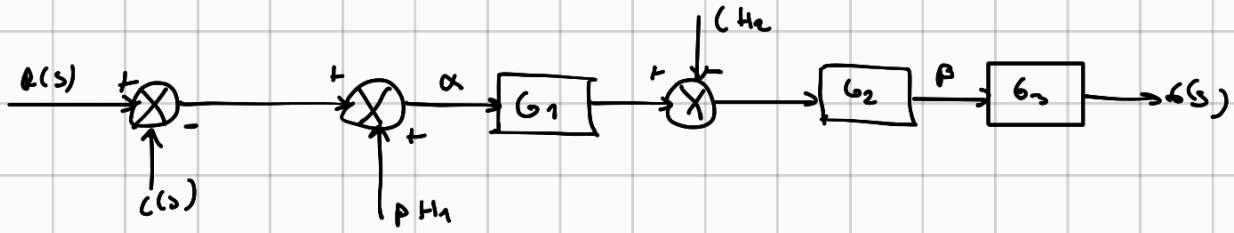
$$TF_2 = \frac{G_1 G_2}{1 - H_1 G_1 G_2}$$

$$TF_3 = \frac{\left(\frac{G_1 G_2}{1 - H_1 G_1 G_2}\right) G_3}{1 + \frac{G_1 G_2}{1 - G_1 G_2 H_1} G_3 \frac{H_2}{G_1}}$$



$$\frac{C(s)}{R(s)} = \frac{TF_3}{1 + TF_3}$$

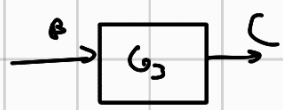
Transverse signal approach



$$\left[\underbrace{(R - C + \beta H_1)}_{\alpha} G_1 - C H_2 \right] G_2 = \beta$$

$$\left[(R - C)\, G_1 - C H_2 \right] G_2 + \beta H_1 G_1 G_2 = \beta$$

$$\beta = \frac{[(R-C)G_1 - CH_2]G_2}{1 - H_1 G_1 G_2}$$

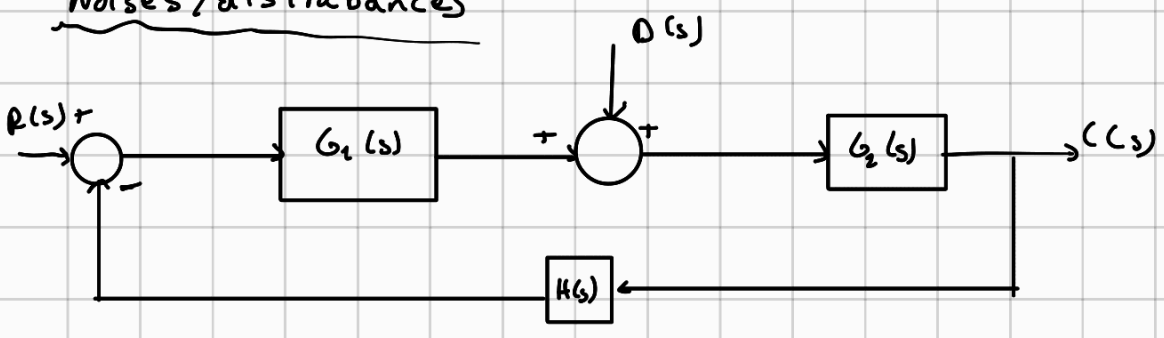


$$C = \beta G_3 \hookrightarrow \frac{[(R-C)G_1 - CH_2]G_2 G_3}{1 - H_1 G_1 G_2} = C$$

$$\Rightarrow \frac{R G_2 G_3}{1 - H_1 G_1 G_2} - \frac{C G_1 G_2 G_3 + C H_2 G_2 G_3}{1 - H_1 G_1 G_2} = C$$

$$C\underbrace{\left[1 + \frac{G_2 G_3}{1 - H_1 G_1 G_2}(G_1 + H_2)\right]}_{q_1} = \underbrace{\left(\frac{G_2 G_3}{1 - H_1 G_1 G_2}\right)}_{q_2} R \rightarrow \frac{C}{R} = \frac{\Phi_2}{\Phi_1}$$

Noises / disturbances



Superposition principle

$$C(s) = C_R(s) + C_D(s)$$

$C_R(s)$: Response due to $R(s)$

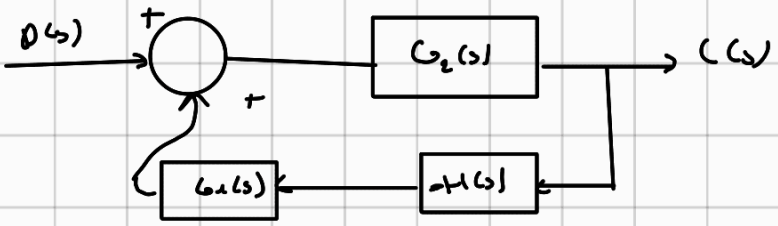
$C_D(s)$: Response due to $D(s)$

case 1)

$R(s) \neq 0$, $D(s) = 0$

$$\frac{C(s)}{R(s)} = \frac{G_1(s) G_2(s)}{1 + G_1 G_2 H} \rightarrow C_R(s) = \frac{G_1 G_2}{1 + G_1 G_2 H} R$$

case 2) $R(s) = 0$, $D(s) \neq 0$



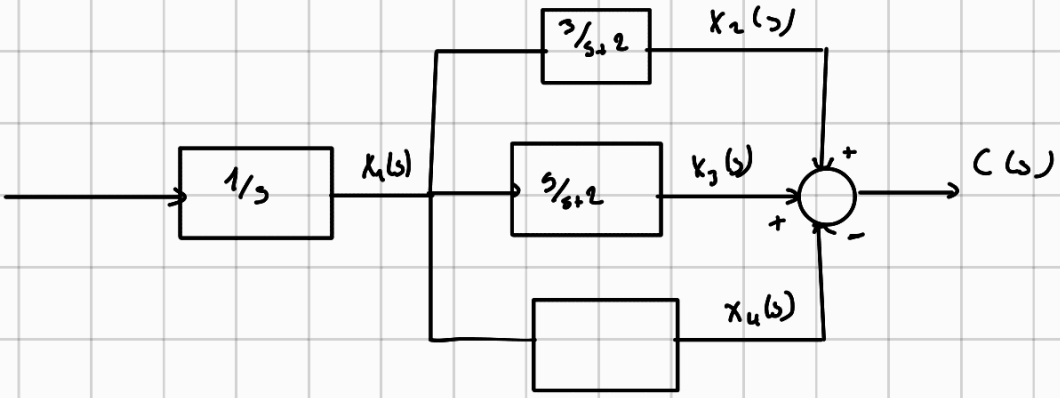
$$\frac{C(s)}{D(s)} = \frac{G_2}{1+G_2G_1H}$$

$$\rightarrow C_D = \left(\frac{G_2}{1+G_1G_2H}\right) D(s)$$

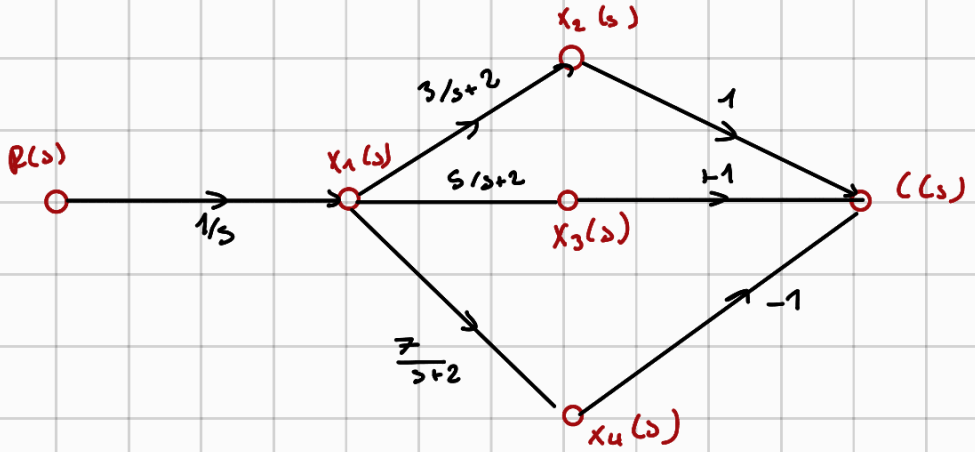
$$C(s) = \frac{G_1G_2}{1+G_1G_2H} R + \frac{G_2}{1+G_1G_2H} D$$

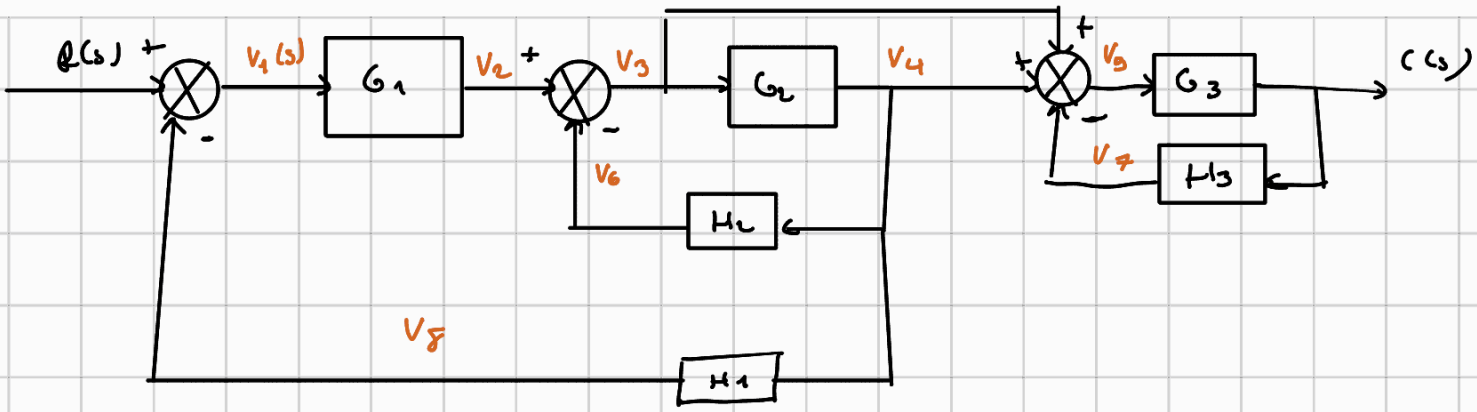
$$= \frac{G_2}{1+G_1G_2H} (G_1R + D)$$

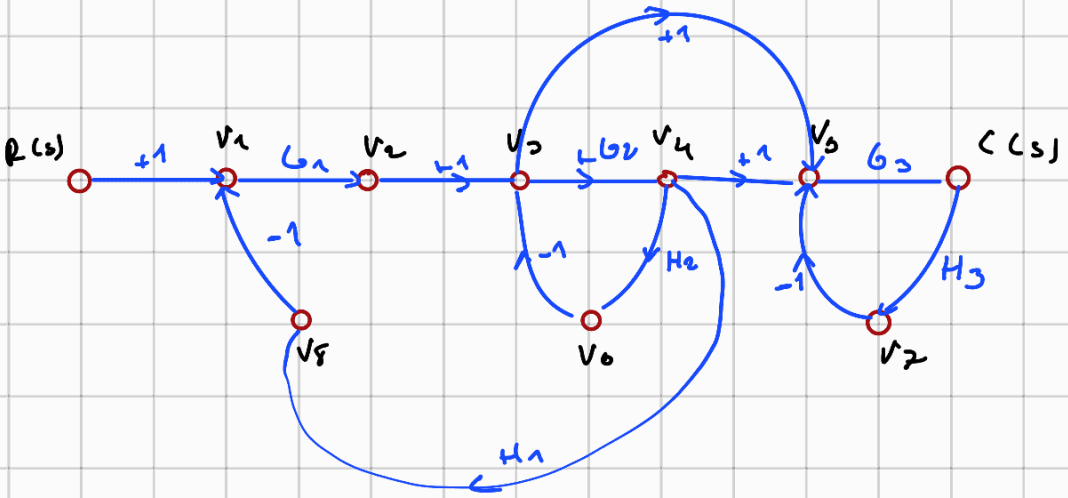
Signal Flow repesentation / graph



6 nodes





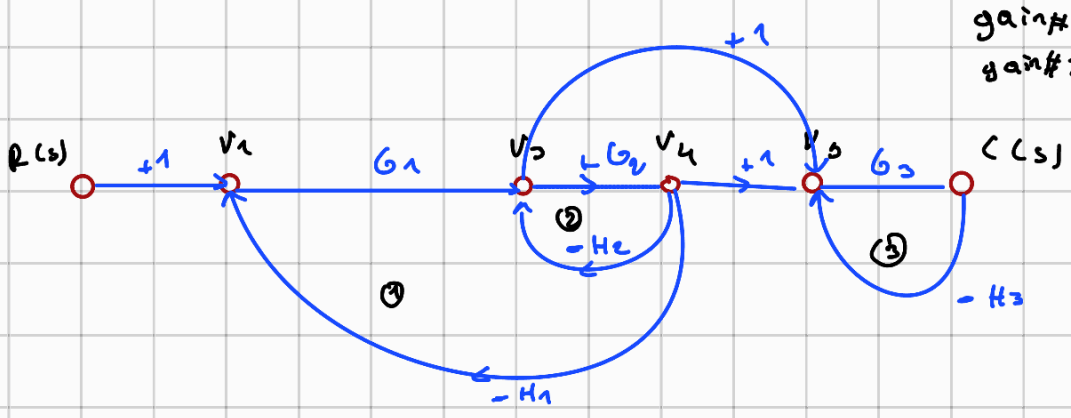


non touching loops (2 ve fazlası)

Birbirine dokunmayan loopların gainleri (küçük loopgainleri çarpılır)

gain#1: $G_2 H_2 G_3 H_3$

gain#2: $G_1 G_2 G_3 H_1 H_3$



There is 3 loop

Feed forward path

R(s) → C(s)

T_1 = # 1 : $G_1 G_2 G_3$ (first path)

T_2 = # 2 : $G_1 G_3$ (second path)

loop

1 : $-G_1 G_2 H_1$

2 : $-G_2 H_2$

3 : $-G_3 H_3$

Mason's rule

$$G(s) = \frac{C(s)}{R(s)} = \frac{\sum_{k=1} T_k \Delta_k}{\Delta}$$

k : number of forward paths

T_k : forward path gain for path # k

$\Delta = 1 - \Sigma$ loop gains $+ \Sigma$ non touching loop gains taken 2 at a time $- \Sigma$ non touching loop gains taken 3 at a time

$+ \Sigma$ non touching loop gains taken 4 at a time ——

$\Delta_k = \Delta - \Sigma$ loop gains that touch the kth forward path

$$\Delta = 1 + G_1 G_2 H_1 + G_2 H_2 + G_3 H_3 + (G_2 G_3 H_2 H_3 + G_1 G_2 G_3 H_1 H_3)$$

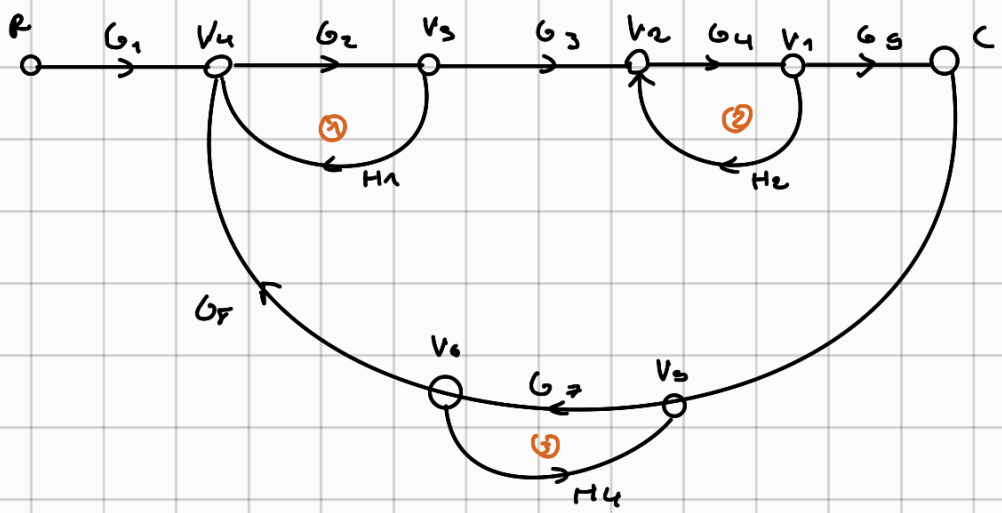
$\Delta_k = \Delta - \Sigma$ loop gains that touch the kth forward path

(make the touching gain $= 0$) $\Rightarrow \Delta$

$\Delta_1 = 1$ $\quad (H_1, H_2, H_3 = 0)$

$\Delta_2 = 1$

$$\frac{C(s)}{R(s)} = \frac{T_1 \Delta_1 + T_2 \Delta_2}{\Delta} = \frac{G_1 G_2 G_3 + G_1 G_3}{\Delta}$$



Feed Forward path : $G_1 G_2 G_3 G_4 G_5$

loops : # 1 : $G_2 H_1$

2 : $G_4 H_2$

3 : $G_2 G_3 G_4 G_5 G_6 G_7 G_F$

4 : $G_7 H_4$

Non touching loop gains (2 ta konata ine)

loops 1,2 : $G_2 H_1 G_4 H_2$

loops 1,4 : $G_2 H_1 G_7 H_4$

loops 2,4 : $G_4 H_2 G_7 H_4$

Non touching loop gains (3 ta konata ine)

loops 1,2,4 : $G_2 H_1 G_4 H_2 G_7 H_4$

$$\Delta = 1 - G_2 H_1 - G_4 H_2 - G_2 G_3 G_4 G_5 G_6 G_7 G_8 - G_7 H_4 + G_2 H_1 G_4 H_2 + G_2 H_1 G_7 H_4 + G_4 H_2 G_7 H_4 - G_2 G_4 G_7 H_1 H_2 H_4$$

$$\frac{C(s)}{R(s)} = \frac{\sum T_k \Delta_k}{\Delta} = \frac{T_1 \Delta_1}{\Delta}$$

$$\Delta_1 = 1 - G_7 H_4$$

